Superstore Sales Analysis

Pandas and Matplotlib analysis covering data extraction, filtering, transformation, KPI calculation, aggregation and visualization.

1. Data Loading and Preparation

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("Downloads/Sample_Superstore.csv",
                 encoding="latin-1")

df["Order_Date"] = pd.to_datetime(df["Order_Date"])
df["Year"] = df["Order_Date"].dt.year
df["Month"] = df["Order_Date"].dt.month
df["Month Name"] = df["Order_Date"].dt.month_name()
df["Month Year"] = df["Order_Date"].dt.to_period("M").astype(str)
```

Explanation: The CSV file is loaded into a DataFrame using Pandas. Order_Date is converted to datetime so year and month attributes can be extracted for analysis. The derived columns make filtering and grouping easier.

2. Filtering and Data Extraction

```
filtered_data = df[
    (df["Year"] == 2016) &
    (df["Segment"] == "Consumer") &
    (df["Region"] == "South") &
    (df["Ship_Mode"] == "Same Day")
]

filtered_data
```

Explanation: Multiple Boolean conditions are combined with & to extract only the records that match the selected year, segment, region and shipping mode.

3. KPI Summary

KPI	Value	Calculation / Meaning
Total Sales	2,297,200.86	Sum of Sales
Total Profit	286,397.02	Sum of Profit
Profit Margin	12.47%	Total Profit / Total Sales × 100
Total Quantity	37,873	Sum of Quantity
Total Orders	5,009	Unique Order_ID count
Average Order Value	458.61	Total Sales / Total Orders

KPI code:

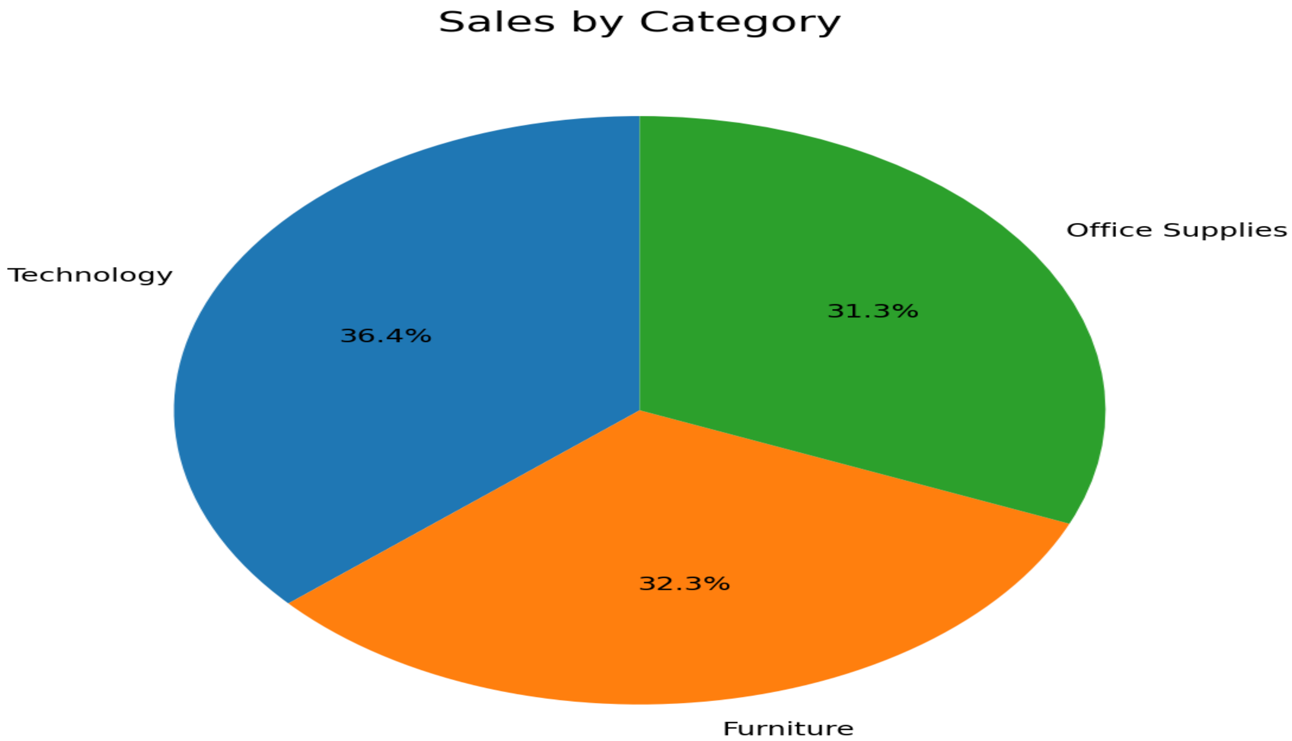
```
total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
total_quantity = df["Quantity"].sum()
total_orders = df["Order_ID"].nunique()
profit_margin = (total_profit / total_sales) * 100
aov = total_sales / total_orders
```

4. Sales by Category

A category-level aggregation is used to understand the contribution of Furniture, Office Supplies and Technology to total sales. The pie chart displays the percentage share of each category.

```
category_sales = (
    df.groupby("Category")["Sales"]
      .sum()
      .sort_values(ascending=False)
)

plt.figure(figsize=(7, 7))
plt.pie(category_sales.values,
        labels=category_sales.index,
        autopct="%1.1f%%",
        startangle=90)
plt.title("Sales by Category")
plt.show()
```



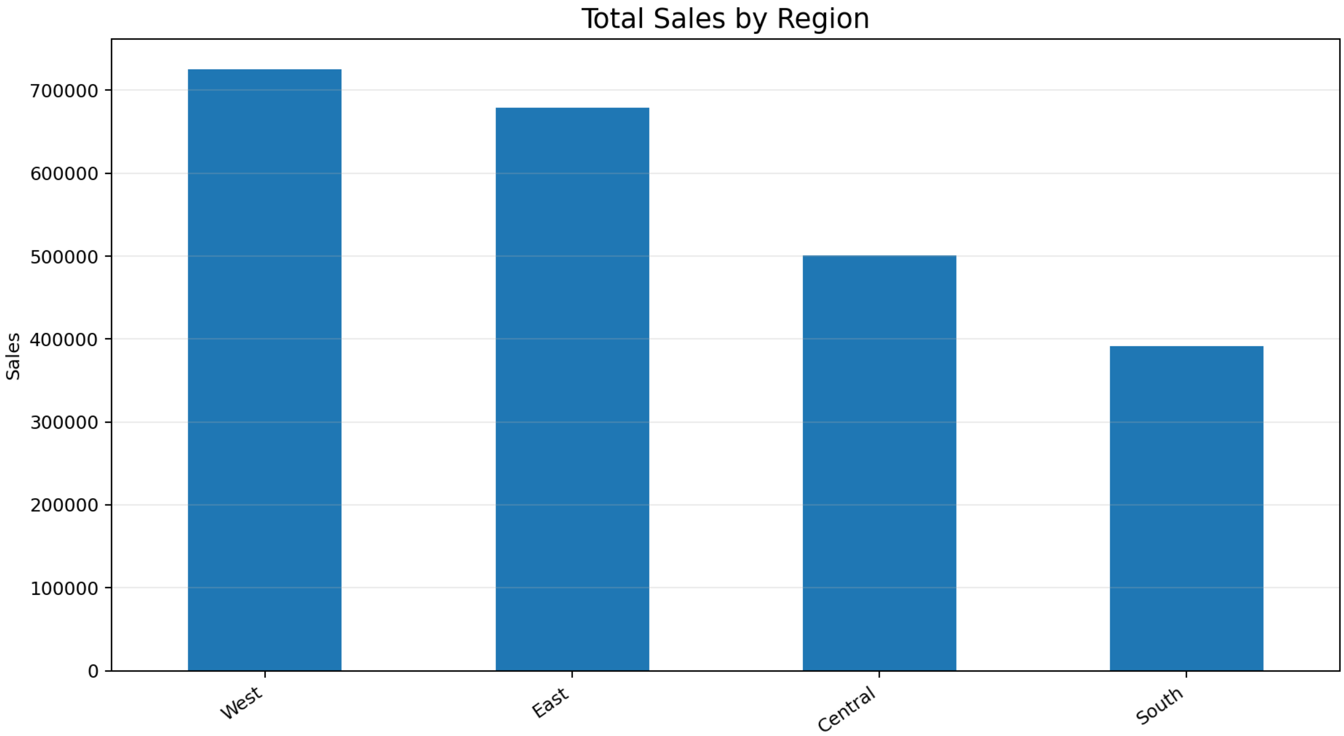
Category	Sales	Share
Technology	836,154.03	36.40%
Furniture	741,999.80	32.30%
Office Supplies	719,047.03	31.30%

5. Total Sales by Region

The Region field is grouped and Sales are summed. West is the highest-performing region, followed by East, Central and South.

```
region_sales = (
    df.groupby("Region")["Sales"]
      .sum()
      .sort_values(ascending=False)
)

plt.figure(figsize=(10, 6))
plt.bar(region_sales.index, region_sales.values)
plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
```



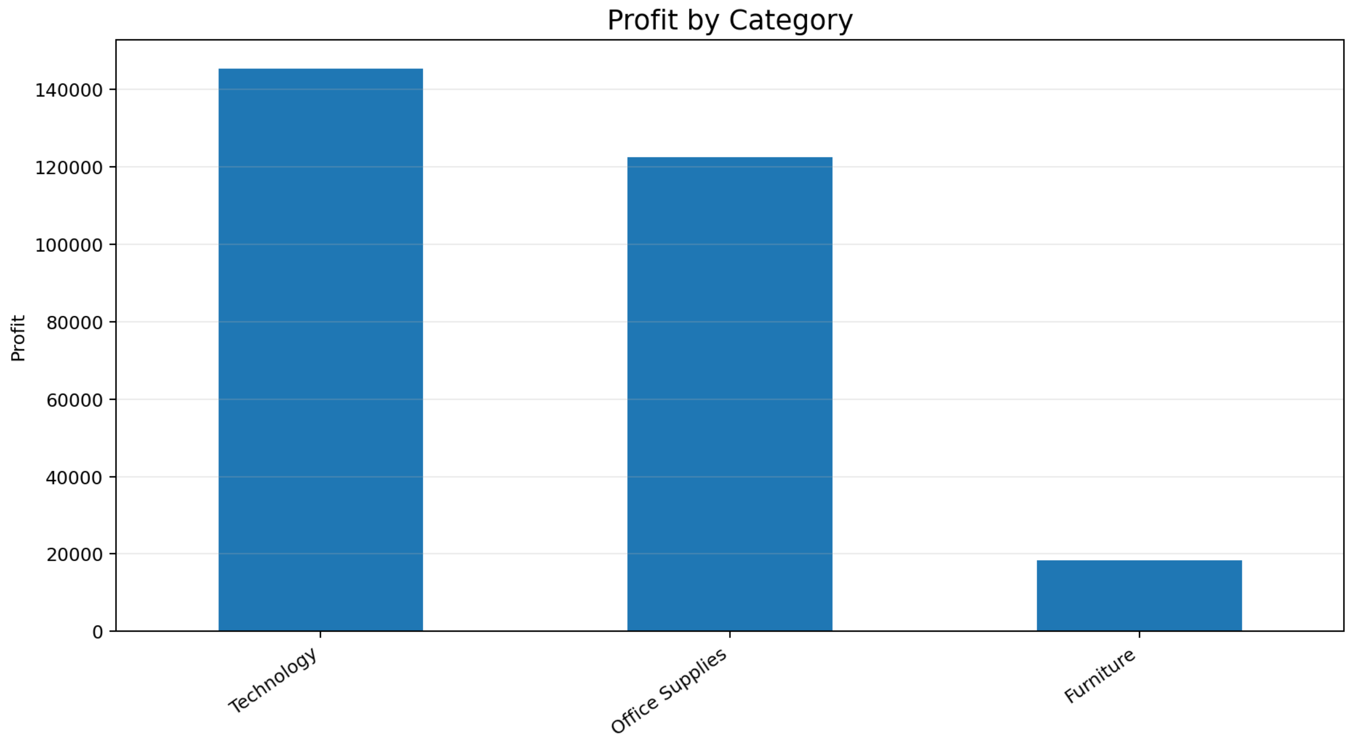
Region	Sales
West	725,457.82
East	678,781.24
Central	501,239.89
South	391,721.91

6. Profit by Category

Sales and Profit are aggregated by Category. Technology produces the highest profit, while Furniture has a much lower profit compared with its sales volume.

```
profit_category = (
    df.groupby("Category")
      .agg(Sales=("Sales","sum"),
           Profit=("Profit","sum"))
      .reset_index()
)

profit_category["Profit Margin %"] = (
    profit_category["Profit"] / profit_category["Sales"] * 100
)
```



Category	Sales	Profit	Profit Margin
Technology	836,154.03	145,454.95	17.40%
Furniture	741,999.80	18,451.27	2.49%
Office Supplies	719,047.03	122,490.80	17.04%

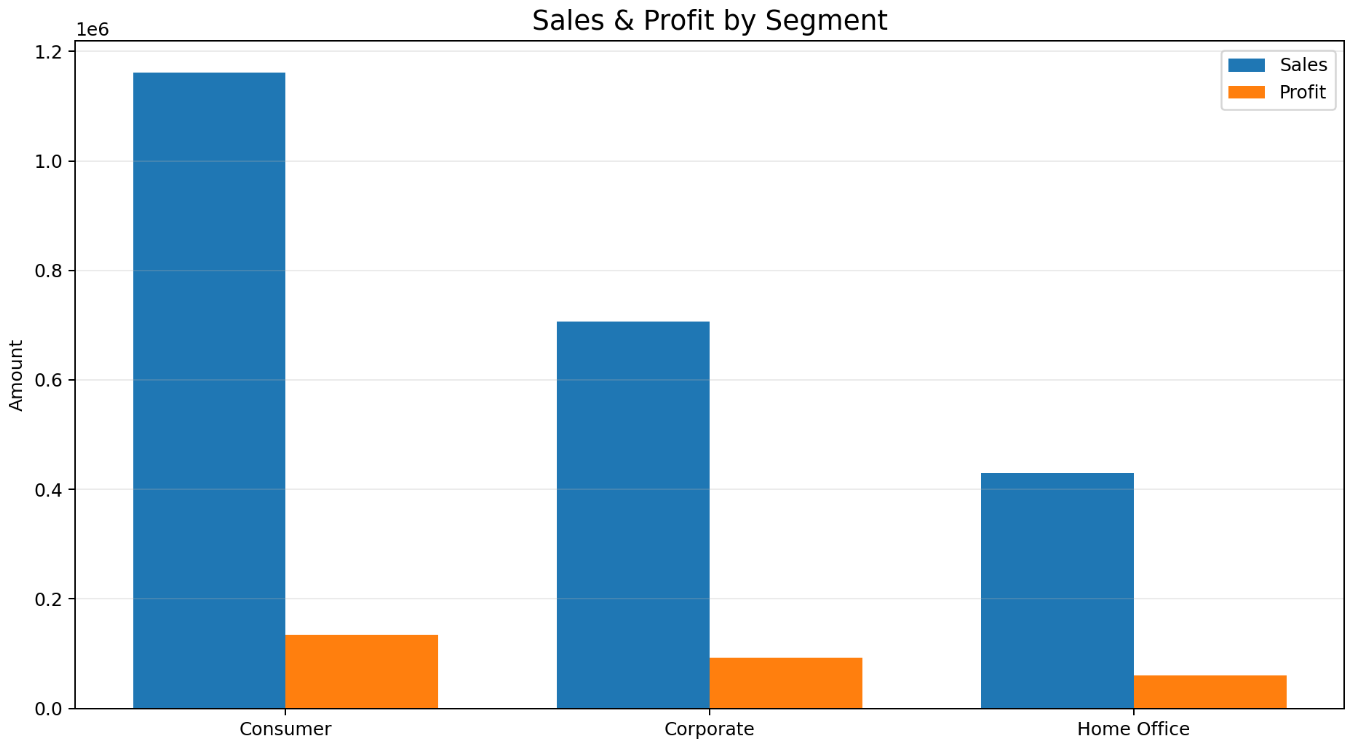
7. Sales and Profit by Segment

The grouped bar chart compares Sales and Profit for Consumer, Corporate and Home Office segments. Consumer contributes the largest sales and profit totals.

```
segment_data = (
    df.groupby("Segment")[["Sales","Profit"]]
      .sum()
)

x = range(len(segment_data))
width = 0.35

plt.figure(figsize=(10, 6))
plt.bar([i-width/2 for i in x], segment_data["Sales"],
        width=width, label="Sales")
plt.bar([i+width/2 for i in x], segment_data["Profit"],
        width=width, label="Profit")
plt.xticks(list(x), segment_data.index)
plt.xlabel("Segment")
plt.ylabel("Amount")
plt.title("Sales & Profit by Segment")
plt.legend()
plt.tight_layout()
plt.show()
```



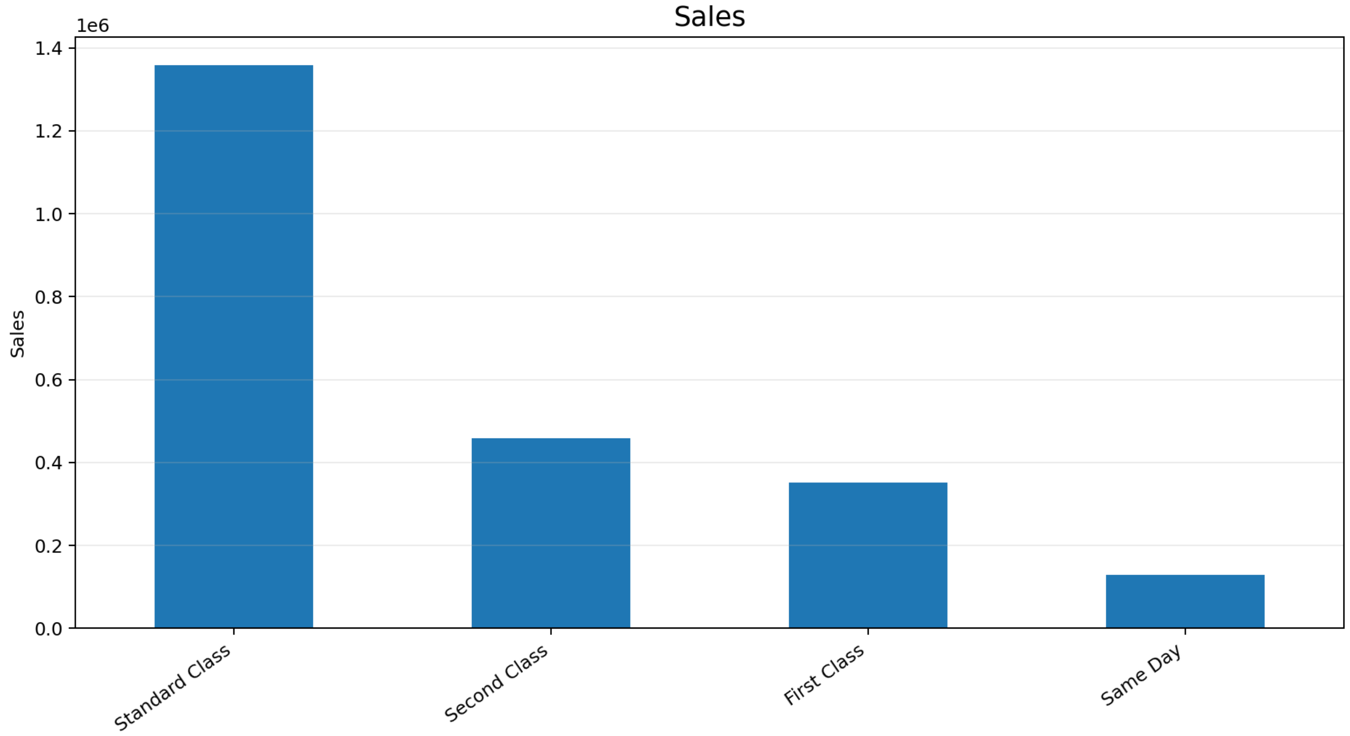
Segment	Sales	Profit
Consumer	1,161,401.35	134,119.21
Corporate	706,146.37	91,979.13
Home Office	429,653.15	60,298.68

8. Sales by Ship Mode

Shipping performance is summarized by Ship_Mode. Standard Class accounts for the largest sales value, while Same Day has the smallest sales value among the four modes.

```
ship_mode_sales = (
    df.groupby("Ship_Mode")["Sales"]
      .sum()
      .sort_values(ascending=False)
)

plt.figure(figsize=(10, 6))
plt.bar(ship_mode_sales.index, ship_mode_sales.values)
plt.title("Sales by Ship Mode")
plt.xlabel("Ship Mode")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
```



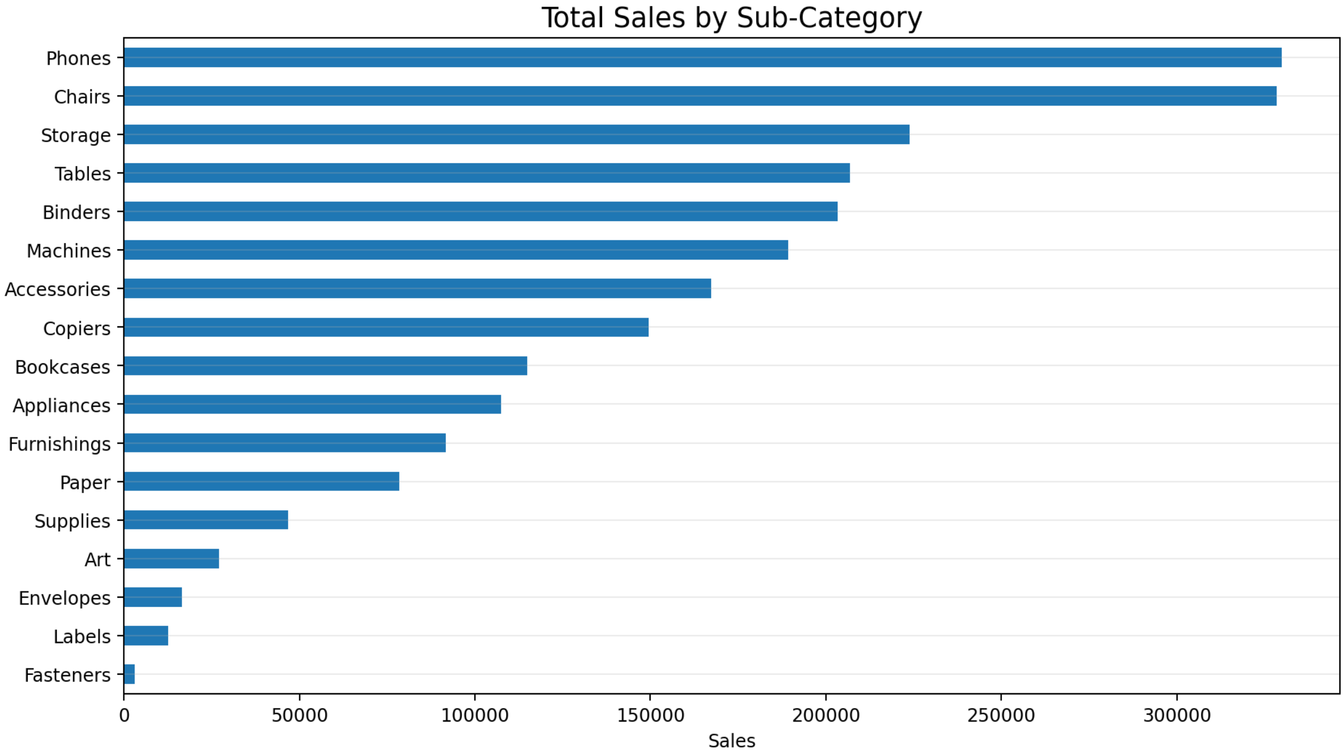
Ship Mode	Sales
Standard Class	1,358,216.00
Second Class	459,193.60
First Class	351,428.40
Same Day	128,363.10

9. Sales by Sub-Category

Sub-category aggregation provides a detailed product-level view. Phones and Chairs are the strongest sub-categories by sales, whereas Fasteners, Labels and Envelopes are the smallest contributors.

```
subcategory_sales = (
    df.groupby("Sub-Category")["Sales"]
      .sum()
      .sort_values(ascending=False)
)

plt.figure(figsize=(10, 7))
plt.barh(subcategory_sales.index, subcategory_sales.values)
plt.title("Total Sales by Sub-Category")
plt.xlabel("Sales")
plt.ylabel("Sub-Category")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



Sub-Category	Sales
Phones	330,007.05
Chairs	328,449.10
Storage	223,843.61
Tables	206,965.53
Binders	203,412.73
Machines	189,238.63
Accessories	167,380.32
Copiers	149,528.03
Bookcases	114,880.00
Appliances	107,532.16
Furnishings	91,705.16
Paper	78,479.21
Supplies	46,673.54
Art	27,118.79
Envelopes	16,476.40
Labels	12,486.31
Fasteners	3,024.28

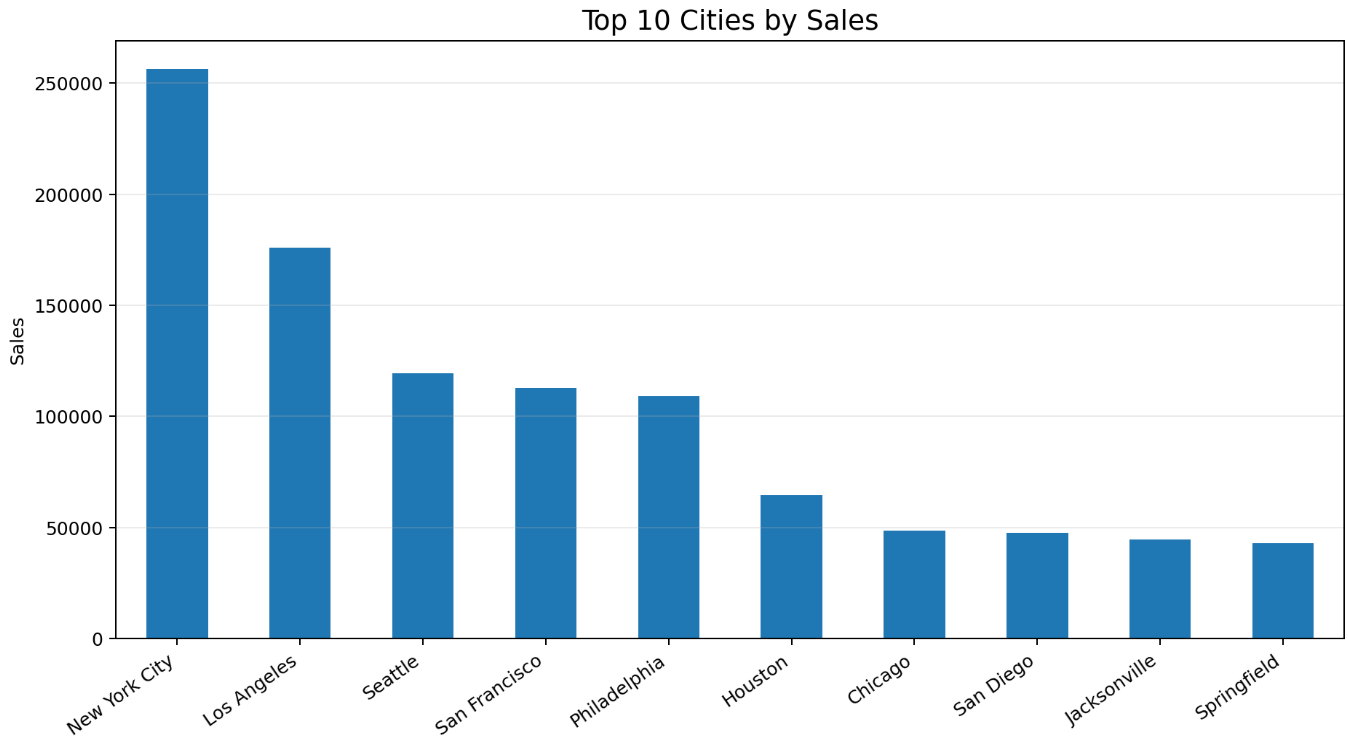
10. Top 10 Cities by Sales

City-level sales are grouped, sorted in descending order and limited to the first ten records. New York City is the leading city in the extracted list.

```
city_sales = (
    df.groupby("City")["Sales"]
      .sum()
      .sort_values(ascending=False)
)

top_cities = city_sales.head(10)

plt.figure(figsize=(10, 6))
plt.bar(top_cities.index, top_cities.values)
plt.title("Top 10 Cities by Sales")
plt.xlabel("City")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



City	Sales
New York City	256,368.16
Los Angeles	175,851.34
Seattle	119,548.74
San Francisco	112,669.89
Philadelphia	109,077.01
Houston	64,504.76
Chicago	48,539.54
San Diego	47,521.03
Jacksonville	44,713.18
Springfield	43,054.34

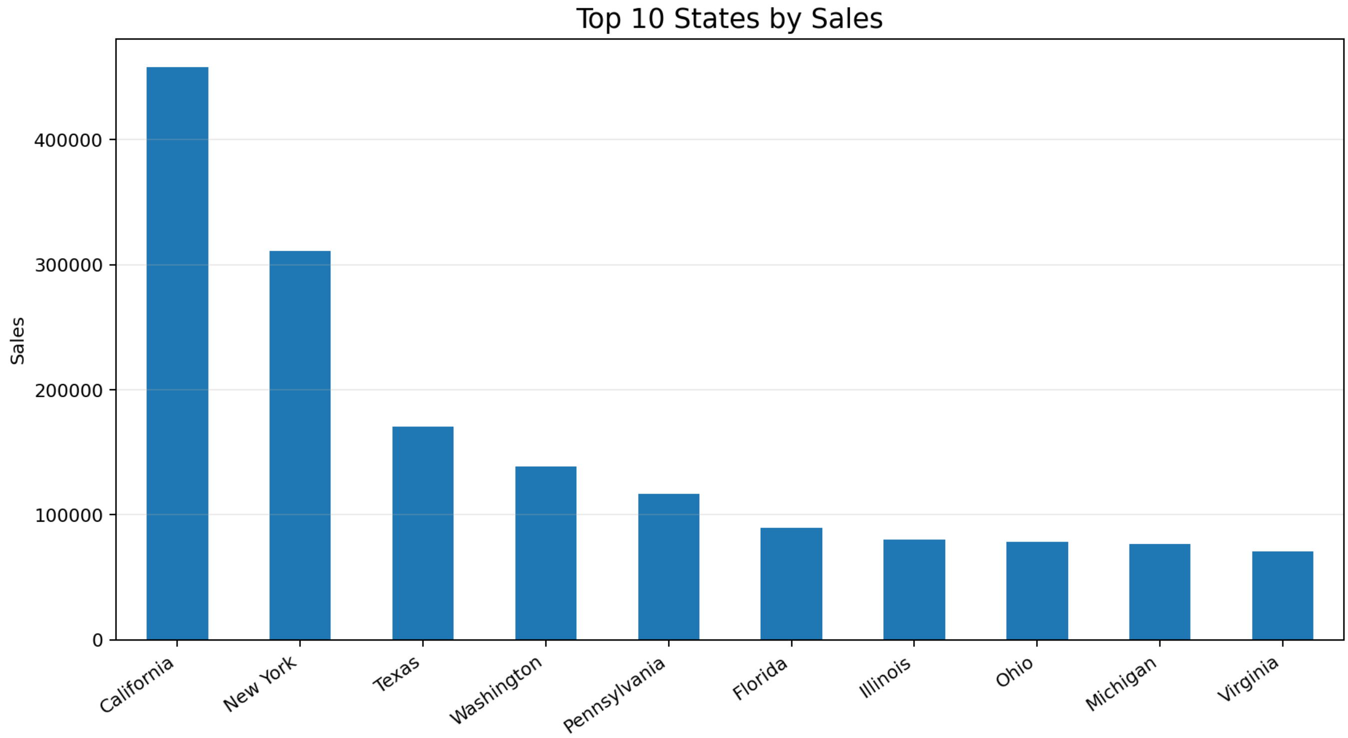
11. Top 10 States by Sales

State and Region are grouped together, Sales are summed and the results are sorted in descending order. California has the highest sales among the top ten states.

```
state_sales = (
    df.groupby(["State","Region"])["Sales"]
      .sum()
      .reset_index()
      .sort_values("Sales", ascending=False)
)

top_states = state_sales.head(10)

plt.figure(figsize=(10, 6))
plt.bar(top_states["State"], top_states["Sales"])
plt.title("Top 10 States by Sales")
plt.xlabel("State")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Rank	State	Region	Sales
1	California	West	457,687.63
2	New York	East	310,876.27
3	Texas	Central	170,188.05
4	Washington	West	138,641.27
5	Pennsylvania	East	116,511.91
6	Florida	South	89,473.71
7	Illinois	Central	80,166.10
8	Ohio	East	78,258.14
9	Michigan	Central	76,269.61
10	Virginia	South	70,636.72

12. Top 10 Products and Final Dataset

The top products are obtained by grouping Product_Name, summing Sales and selecting the first ten sorted results.

Rank	Product	Sales
1	Canon imageCLASS 2200 Advanced Copier	61,599.82
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27,453.38
3	Cisco TelePresence System EX90 Videoconferencing Unit	22,638.48
4	HON 5400 Series Task Chairs for Big and Tall	21,870.58
5	GBC DocuBind TL300 Electric Binding System	19,823.48
6	GBC Ibimaster 500 Manual ProClick Binding System	19,024.50
7	Hewlett Packard LaserJet 3310 Copier	18,839.69
8	HP Designjet T520 Inkjet Large Format Printer - 24" Color	18,374.90
9	GBC DocuBind P400 Electric Binding System	17,965.07
10	High Speed Automatic Electric Letter Opener	17,030.31

13. Export and Final Analytical Dataset

```
df["Profit Margin %"] = (
    df["Profit"] / df["Sales"] * 100
).round(2)

df.to_csv("Superstore_Cleaned.csv", index=False)

final_data = df[
    ["Order_ID","Order_Date","Year","Month Name",
     "Segment","Region","State","Category",
     "Sub_Category","Ship_Mode","Sales","Profit",
     "Profit Margin %","Quantity","Product_Name"]
]
```

Explanation: The cleaned dataset is exported as Superstore_Cleaned.csv. The final_data DataFrame keeps the fields required for reporting, visualization and dashboard development.

Conclusion

Extract → Filter → Transform → Calculate → Aggregate → Visualize → Export. The analysis provides a complete view of Superstore performance across categories, regions, segments, shipping modes, sub-categories, cities, states and products. The unsuccessful/error output is intentionally not included.